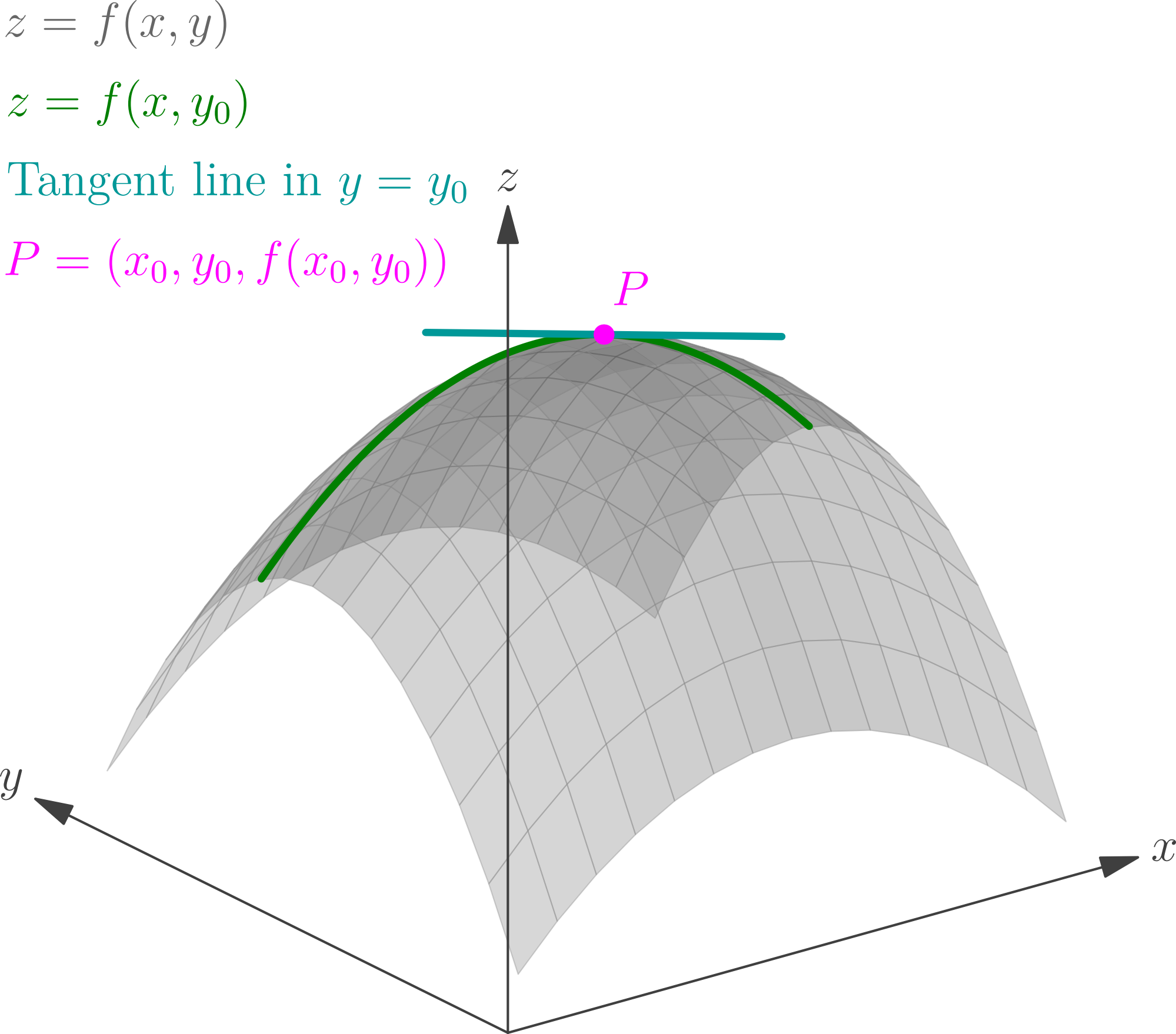
$z = f(x, y)$

$z = f(x, y_0)$

Tangent line in $y = y_0$

$P = (x_0, y_0, f(x_0, y_0))$